



Climate Change

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Abstract

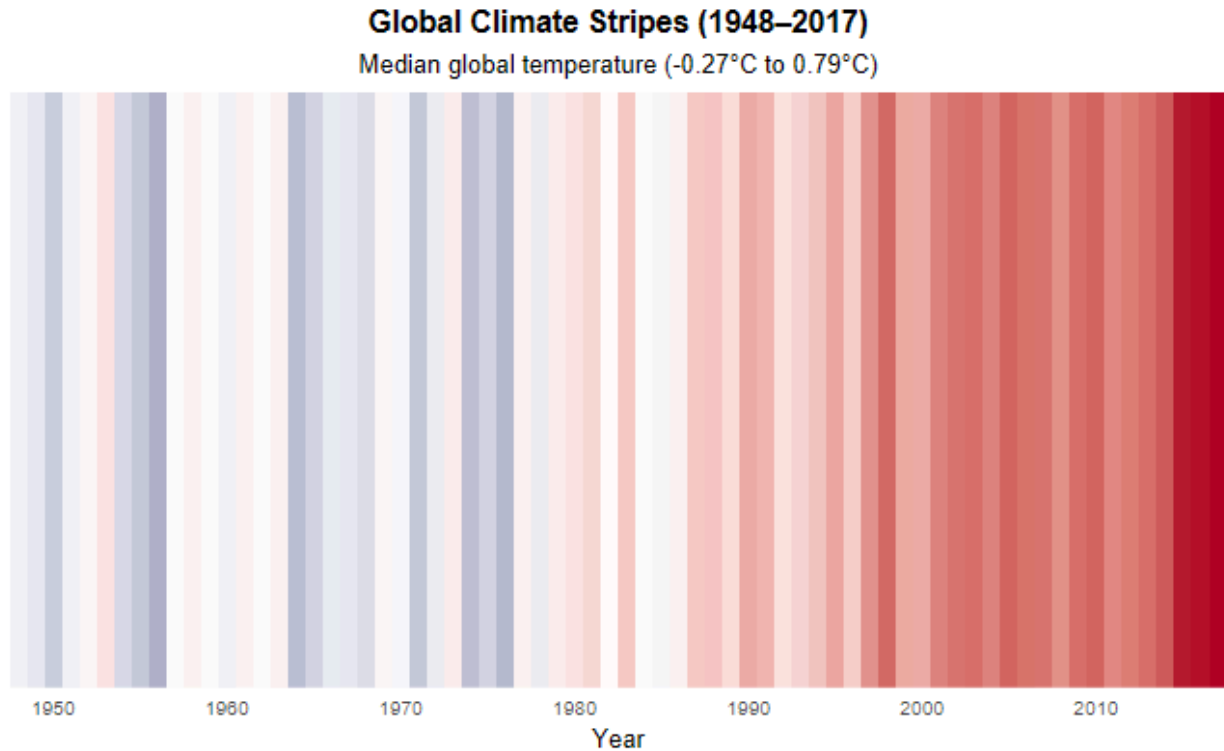
Climate change is one of the most urgent global challenges of the 21st century, driven primarily by rising greenhouse gas emissions from human activity. This project explores the relationship between carbon dioxide (CO₂) emissions, global temperature change, and economic development through data-driven visual storytelling. Using internationally comparable data from Our World in Data, the analysis is structured around three key insights: the link between rising CO₂ emissions and global warming, the role of economic growth in shaping emissions patterns, and the unequal distribution of historical climate responsibility across countries. Through visualizations such as climate stripes, emissions–temperature relationships, and GDP–CO₂ comparisons, the project highlights how industrialization, income, and policy choices influence emission trajectories. The findings demonstrate that while emissions are rising globally, responsibility for climate change is uneven, with historically industrialized and wealthier nations contributing disproportionately. Recognizing these disparities is essential for developing fair and effective climate policies that balance mitigation efforts with sustainable development.

1. Introduction

Climate change has become a defining issue of our time, affecting ecosystems, economies, and societies worldwide. Rising global temperatures, increasingly frequent extreme weather events, and long-term environmental degradation are closely linked to the accumulation of greenhouse gases in the atmosphere, particularly carbon dioxide (CO₂). Since the Industrial Revolution, human activities such as fossil fuel combustion, industrial production, and land-use change have significantly increased atmospheric CO₂ concentrations, driving long-term global warming. While climate change is a global phenomenon, its causes and responsibilities are not evenly distributed. Countries differ widely in their levels of economic development, energy use, population size, and historical emissions. As a result, understanding climate change requires more than just looking at current emission totals—it demands a multidimensional perspective that includes historical responsibility, per-capita emissions, and economic context. This project uses data visualization as a storytelling tool to explore three central insights. First, it examines the relationship between global CO₂ emissions and rising global temperatures. Second, it analyzes how economic growth influences emissions patterns across countries. Third, it investigates the unequal distribution of climate responsibility by comparing current, historical, and per-capita emissions. By combining visual evidence with contextual interpretation, the project aims to provide a clearer, more nuanced understanding of the drivers of climate change and the challenges of addressing it fairly.

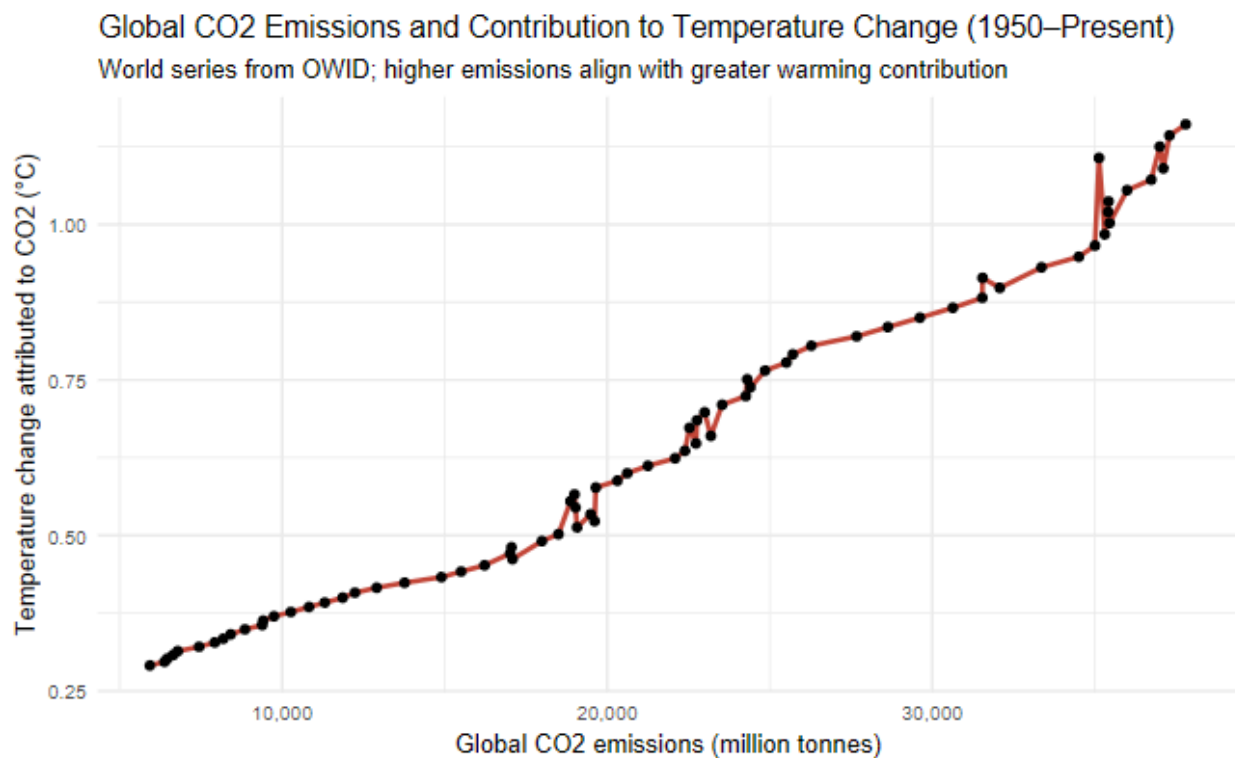
2. Insight 1: Global CO₂ Emissions and Their Contribution to Global Temperature Change

2.1 Global Climate Stripes



This climate-stripes chart visualizes how global temperatures have changed from 1948 to 2017. Each vertical stripe represents one year, with blue tones indicating cooler-than-average years and red tones indicating warmer-than-average years. The progression from predominantly blue stripes in the mid-20th century to deep red stripes in recent decades clearly illustrates the long-term warming trend. This simple but powerful graphic highlights how global temperatures have steadily increased over time, reflecting the accelerating impact of human-driven climate change.

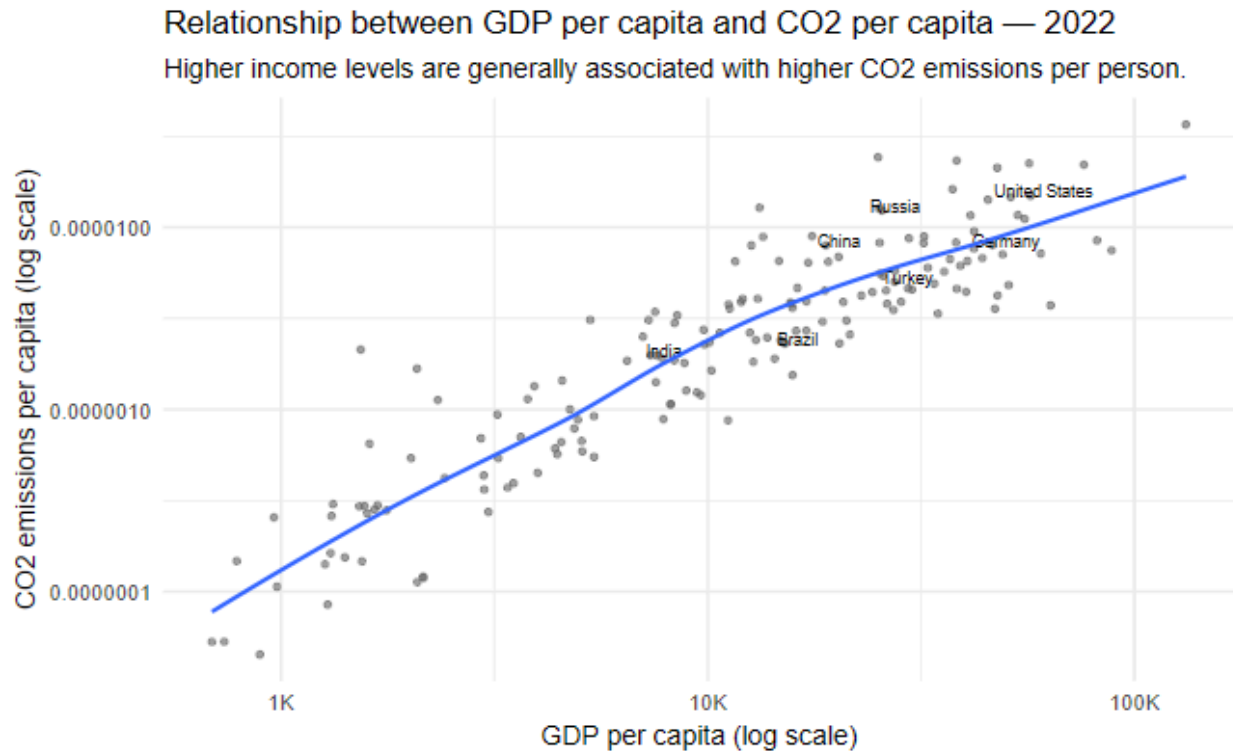
2.2 Global CO₂ Emissions and Their Contribution to Global Temperature Change (1950–Present)



This chart shows how global carbon dioxide (CO₂) emissions have risen since 1950 and how this rise corresponds to an increasing contribution to global temperature change. As the line moves upward and to the right, it illustrates that higher annual global CO₂ emissions are consistently associated with greater warming influence on the Earth's climate. The relationship is not random—temperature impact steadily increases as emissions rise, reflecting the cumulative nature of CO₂ in the atmosphere. This visualization highlights the close connection between human-driven emissions and global warming, emphasizing that continued emission increases lead to progressively stronger climate effects.

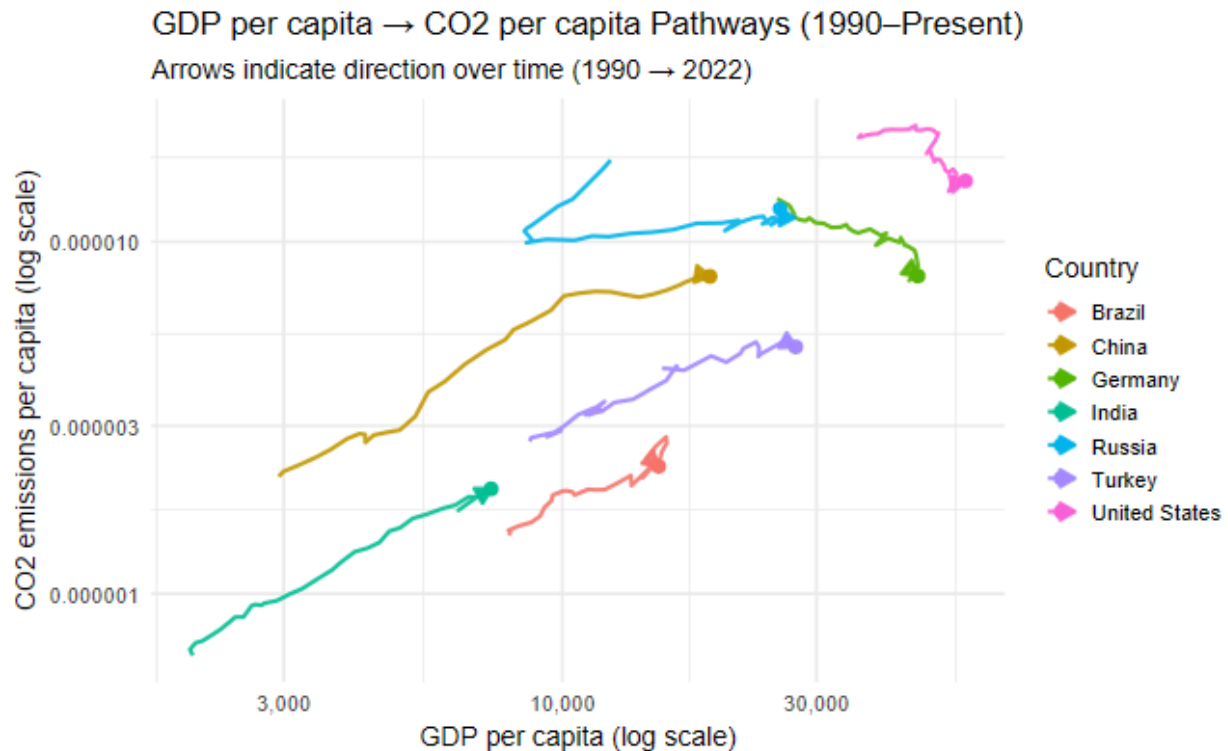
3. Insight 2: Economic Growth Shapes CO₂ Emissions

3.1 GDP per capita vs. CO₂ per capita



This chart shows that countries with higher income levels tend to emit more CO₂ per person. Each point represents a country, and both axes use a log scale to allow comparison across very different economies. The upward trend indicates that as GDP per capita increases, CO₂ emissions per capita also generally rise. However, the points are spread out rather than forming a perfect line, which means income alone does not determine emissions. Some wealthy countries emit less because they use cleaner energy sources or have strong climate policies, while some rapidly developing countries emit more due to coal-based industrial growth.

3.2 Economic Development and CO₂ Emissions Pathways (1990–Present)



This figure shows how economic development relates to CO₂ emissions over time for selected countries (1990 → present). Each line represents a country's trajectory: as GDP per capita increases (moving right), CO₂ emissions per capita may increase or decrease (moving up or down). The arrows indicate the direction of change over time.

Country & Interpretation

China & India: Rapid economic growth accompanied by rising emissions, reflecting carbon-intensive industrialization.

Turkey & Brazil: Moderate income growth with increasing emissions, indicating continued dependence on fossil fuels.

Russia: High emissions per capita with relatively unstable economic growth, shaped by energy-heavy industries and resource extraction.

United States: Very high income with declining emissions per person, suggesting partial decoupling through cleaner energy and efficiency.

Germany: High income with clear, long-term decline in emissions, showing successful low-carbon transition policies.

Countries do not follow the same development–emissions pathway. Some countries (e.g., Germany, United States) reduce emissions while maintaining high income, while others (e.g., China, Turkey) still experience rising emissions as their economies grow.

4. Insight 3: Climate responsibility is unequal - historical emissions are concentrated in rich countries.

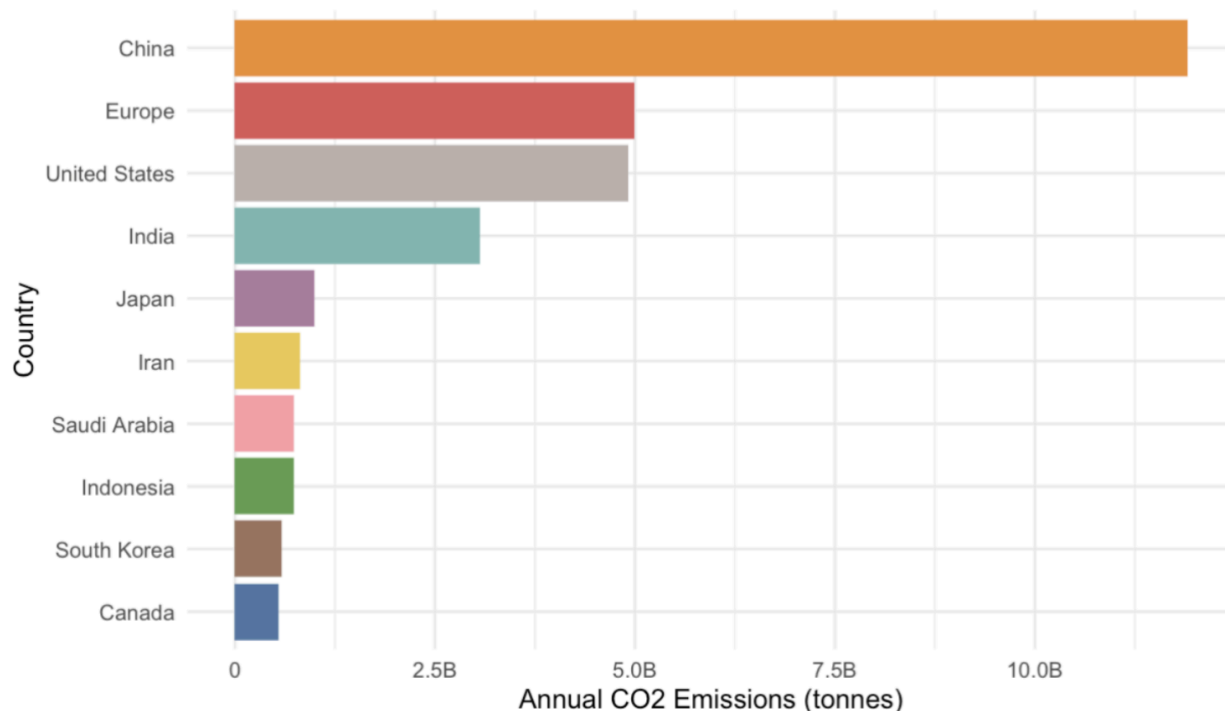
It has become clear that over the last few decades, humans are emitting more carbon dioxide into the earth's atmosphere than ever before and it is warming our planet. While the entire earth warms, not all nations share equal responsibility for this crisis. As we will see, historically, more developed or industrialized countries and continents like the United States and member states of the European Union, and as a whole Europe, have released more carbon dioxide into the atmosphere that has accumulated over the decades. On the other hand, developing countries like India and many in Africa are only now beginning to emit more carbon as they lift themselves into the industrialized world. In addition, they argue that emissions by the developed countries are for "lifestyle" emissions and for the developing are "survival" emissions. This inequality raises the critical question of who should bear the responsibility of climate change?

In order to come up with any kind of answer, we need to break down the question into three parts. First, we first need to identify which countries emit the most carbon dioxide today, then look at which countries have emitted the most in total, then finally identify the countries which emit the most per capita. This will allow us to understand both the present and historical dimensions of responsibility. By comparing total, annual, and per capita emissions, we can form a more balanced picture of climate responsibility.

The data for this section has been collected from Our World In Data (OWID). The dataset provides internationally comparable estimates of carbon dioxide emissions across countries and regions over time. The dataset consists of 4 properties, **Entity** for the country name, **Code** for the associating country code, **Year** and **Annual CO2 Emissions** for the carbon emitted in that year with value in tonnes. [1]

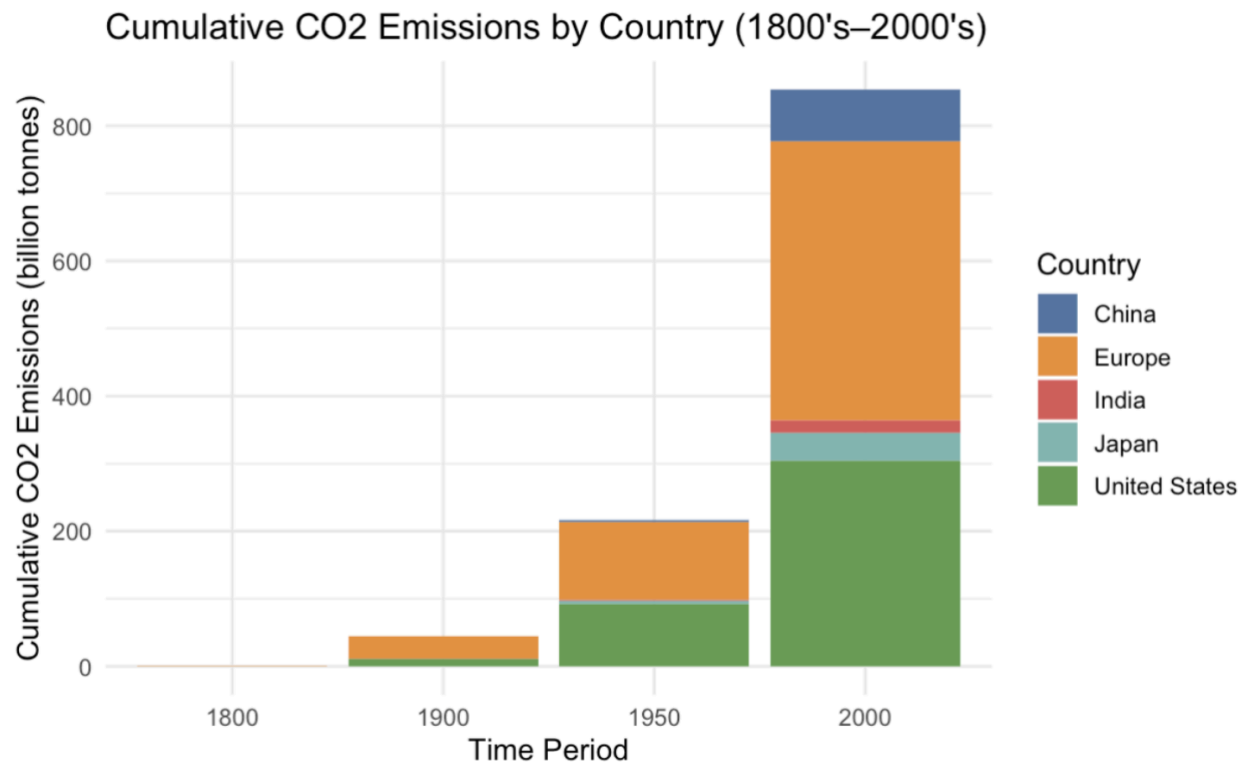
Question 1: Which Countries Emit The Most Carbon Dioxide Today?

Top 10 CO2 Emitting Countries (+ World) in 2023

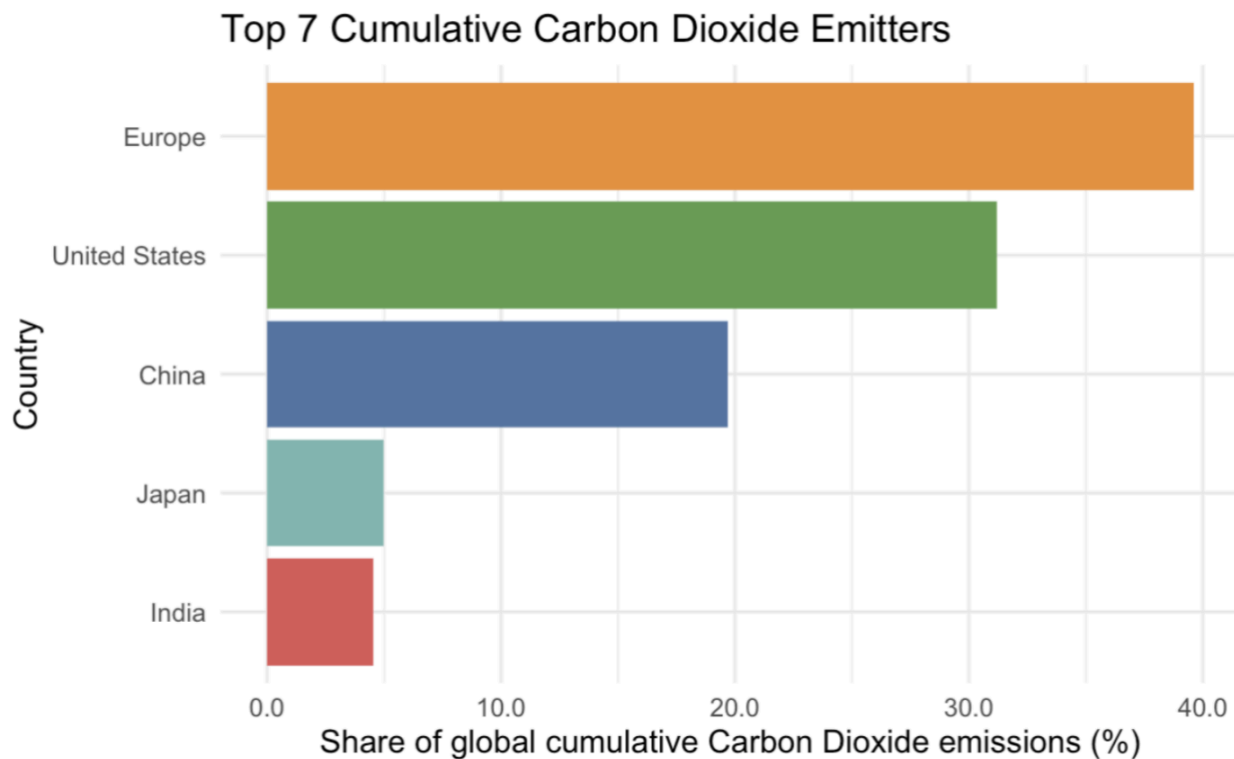


Looking at the top 10 countries that emit the most carbon dioxide in the year 2023, we see that China is by far and away the world's largest emitter with almost 12 billion tonnes of carbon dioxide. It's followed by Europe at almost five billion and not far behind is the United States also at around five billion. Given that the total world combined carbon dioxide emissions in 2023 were 37 billion, and that the top three emitters in total emit 22 billion (almost 60%). It is clear that without the willingness and action of these three industrial giants, fixing climate will remain an impossible task. These three economies hold the greatest leverage in reducing global emissions, not only because of their industrial scale but also their technological and financial capacity to lead the transition toward cleaner energy. Their policies, innovations, and international cooperation will largely determine the success of our collective global climate goals. However, if we only look at the total global emissions today, we are not able to grasp the entire picture of 'responsibility'. Therefore we need to look at historical emissions and how much countries have emitted over time.

Question 2: Which Countries Have Emitted The Most Carbon Dioxide In Total?

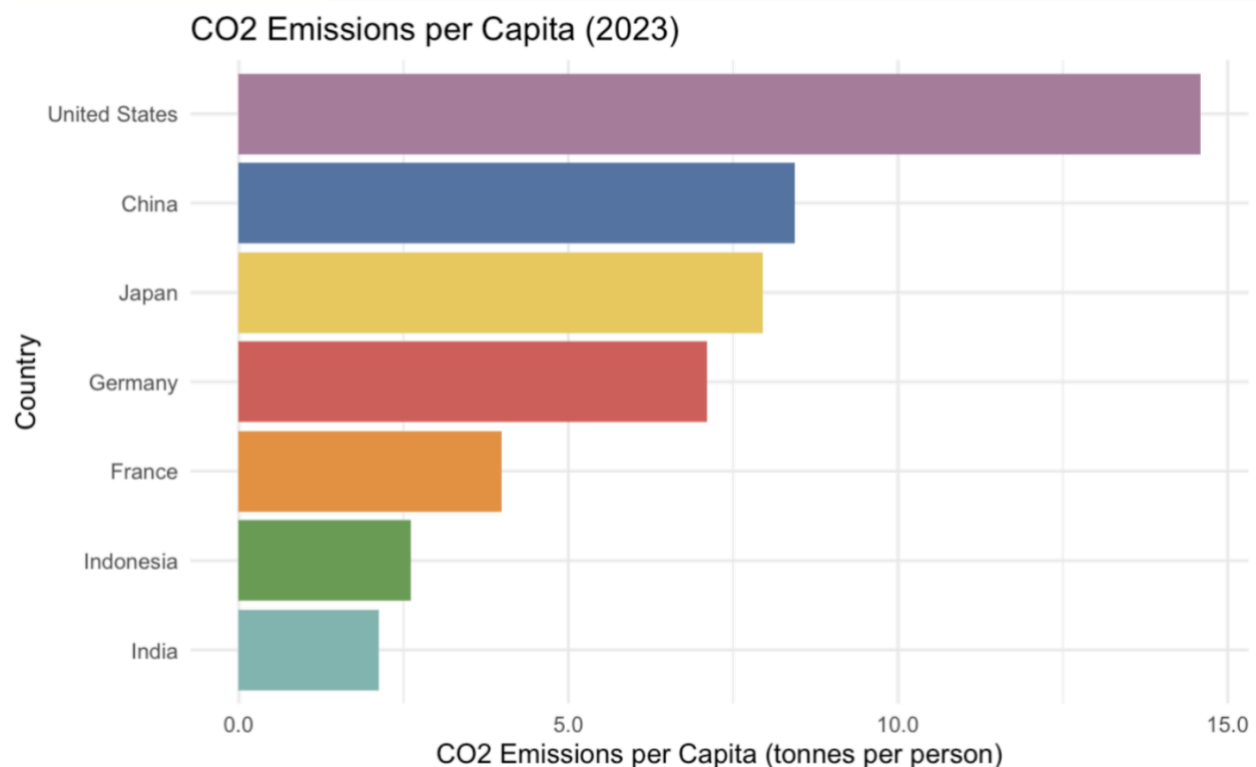


In order to grasp a better understanding about climate responsibility, we need to look at which countries have contributed most to the total atmospheric carbon dioxide burden and how these contributions have accumulated across different historical periods. Starting with the 1800s, we see little to no emissions by the world but there is a slight increase in the 1900s with most of the carbon emissions resulting from Europe and the United States. There is then a sharp increase to the next century with around 250 billion tonnes of carbon dioxide being released into the atmosphere from Europe alone. There is an even sharper increase in the next century as carbon emissions skyrocket even though we are just 23 years into the century at the time of this graph. We observe that Europe is by far and away the biggest emitter of carbon dioxide and the United States is a close second once again. This implies that historical responsibility for climate change lies overwhelmingly with long-industrialized, wealthier regions rather than with developing nations.



If we look at historical emissions of the top emitters, there is a drastic change in the outlook. By far, the United States and Europe have emitted more carbon dioxide into the atmosphere. Europe is responsible for 34% of the world's historical emissions, while The United States is responsible for 27% with most of this coming from the 20th century. Third is China with 17%, which is exactly half of Europe's contribution. Then there is a drastic fall when it comes for countries like India and Japan with both under 5%. The main takeaway from this chart is that historical responsibility for climate change lies overwhelmingly with early-industrialized regions, not with today's developing nations. Even though countries with large populations like China and India, are major emitters today, their cumulative contribution is far smaller because their large-scale industrialization began much later compared to the western world. In contrast, Europe and the United States have been emitting carbon for over a century, and those past emissions have accumulated in the atmosphere, driving much of the warming we see today. Therefore, the narrative that rapid climate change is really the responsibility of the developing world with its current day emission rates is not true.

Question 3: Which Countries Emit The Most Carbon Dioxide Per Person?



When examining carbon dioxide emissions on a per capita basis, the disparities between the countries becomes even clearer. Countries such as the United States, China, and Japan emit far more carbon dioxide per person than larger but less industrialized nations like India or Indonesia. This shows us that while developing countries may contribute more in total due to population size, their citizens individually emit far less. Per capita emissions highlight the lifestyle and consumption patterns that drive climate inequality. Developed economies maintain high energy use, extensive networks, and energy demanding industries that further raise their individual carbon footprints. Meanwhile, developing nations continue to emit primarily for basic growth and infrastructure needs. This brings to light the conversation about responsibility. It is not only about total emissions, but also about how much each person contributes.

We have now looked at which countries contribute the most to global carbon dioxide emissions today in total, per capita and historically. We found that today's total emissions are dominated by large economies such as China, the United States and Europe, while historical emissions are overwhelmingly concentrated in early-industrialized regions, particularly Europe and the United States. If we look at the per-capita basis, wealthier countries emit far more per person than developing nations, reflecting differences in consumption and lifestyle rather than population size alone. A viable solution forward is for developed countries to use resources, skilled workforces,

and carbon capture technology to develop low-cost low-carbon emitting solutions and for developing countries to focus on sustainable development, expand with renewable resources and low-carbon solutions. To conclude, we can say that addressing climate change fairly and effectively requires recognizing the differing responsibilities of countries based on historical contributions and current emissions, with developed nations taking the lead in mitigation while supporting sustainable growth for developing countries.

Conclusion

This project has examined climate change through a data-driven and visual storytelling approach, focusing on the relationships between carbon dioxide emissions, global temperature change, economic development, and climate responsibility. The analysis demonstrates a clear and consistent link between rising global CO₂ emissions and increasing global temperatures, reinforcing the central role of human activity in driving contemporary climate change. The findings further show that economic growth significantly shapes emissions patterns, but countries follow different development–emissions pathways. While some high-income countries have begun to reduce emissions through cleaner energy sources, efficiency improvements, and climate policies, others continue to experience rising emissions as their economies expand. This highlights that economic growth alone does not determine emissions outcomes; policy choices and energy structures play a critical role. Most importantly, the analysis reveals that climate responsibility is deeply unequal. Although current total emissions are dominated by large economies such as China, the United States, and Europe, historical emissions are overwhelmingly concentrated in early-industrialized regions, particularly Europe and the United States. On a per-capita basis, wealthier countries continue to emit significantly more CO₂ per person than developing nations, reflecting differences in consumption patterns and lifestyles rather than population size alone. These findings challenge simplified narratives that place the primary responsibility for climate change on today's developing economies. Overall, addressing climate change in a fair and effective manner requires acknowledging both historical and present-day inequalities in emissions. Developed countries must take the lead in reducing emissions, investing in low-carbon technologies, and supporting global mitigation efforts, while developing countries should be enabled to pursue sustainable growth through renewable energy and low-carbon development pathways. By combining data visualization with contextual interpretation, this project underscores the importance of equity, responsibility, and cooperation in shaping future climate action.

References (APA-7)

- [1] Ritchie, H., Roser, M., & Rosado, P. (2023). *CO₂ and greenhouse gas emissions*. Our World in Data. <https://ourworldindata.org/co2-emissions>
- [2] World Bank. Population, total (Country-Year) [Data set]. <https://data.worldbank.org/>